SOX9 collapse in cholangiocytes: the molecular phenotype of identity loss in primary sclerosing cholangitis

A single-cell transcriptomic exploration of cholangiocyte identity loss, with idiopathic adulthood ductopenia as the motivating disease context.

EXPLORATORY ANALYSIS · NOT A CAUSAL CLAIM · NOT A THERAPEUTIC CLAIM

ABSTRACT

Idiopathic adulthood ductopenia is a rare cholestatic syndrome characterised by progressive loss of intrahepatic bile ducts; no public single-cell IAD datasets exist. Using primary sclerosing cholangitis (PSC) as a tractable cholangiopathy proxy, we analysed 21,038 cholangiocytes pooled from 16 public datasets — 4,092 from PSC donors. The most consistent molecular phenotype of within-disease progression is collapse of SOX9, the core cholangiocyte transcription factor. SOX9-low PSC cholangiocytes show coordinated loss of the canonical biliary-identity program. Candidate upstream pathways remain limited: oxidative stress retains borderline directionality at single-cell resolution; IL6 / STAT3 — the strongest donor-level signal — does not survive cell-level testing. We nominate SOX9-low cholangiocytes as a candidate molecular surrogate for the cholangiocyte-loss endpoint relevant to ductopenia. No causal upstream regulator is identified.

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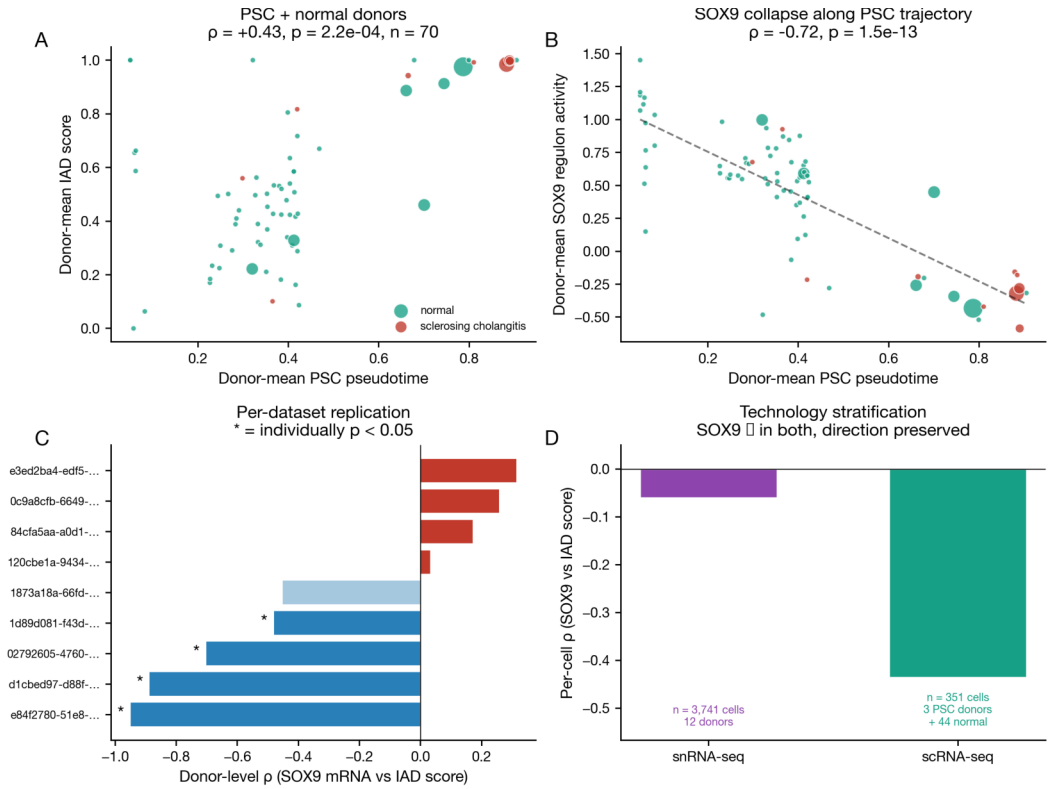
Figure 1 · SOX9 collapse along PSC progression



A Donor-mean PSC pseudotime vs IAD score. $\rho = +0.43$, $p = 2 \times 10^{-4}$, $n = 70$ donors — the diffusion-pseudotime trajectory is consistent with PSC progression.

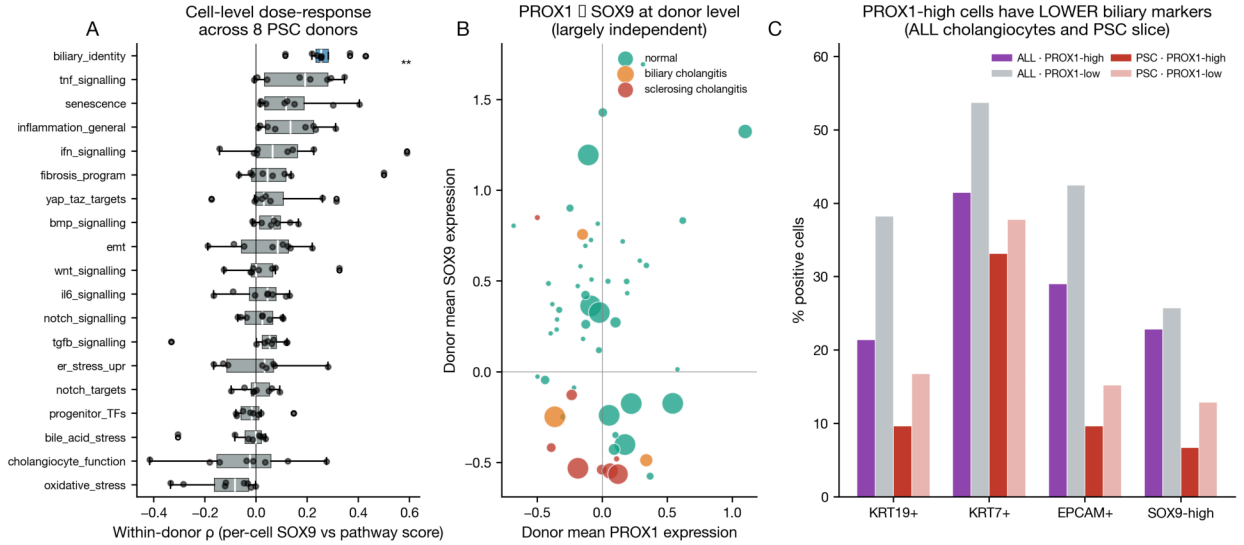
B Same axis vs donor-mean SOX9 regulon activity. $\rho = -0.72$, $p = 1.5 \times 10^{-13}$.

C Per-dataset replication across 9 source datasets. Bars marked * are individually significant at $p < 0.05$; every significant bar points in the predicted negative direction.

D Technology stratification. Per-cell ρ between SOX9 and IAD score is negative in both snRNA-seq and scRNA-seq. snRNA magnitude is attenuated because cytoplasmic mRNAs such as SOX9 are systematically under-detected in nuclei.

> KEY TAKEAWAY
>
> *SOX9 collapse along PSC progression replicates across donors, datasets, and sequencing technologies. This is the most consistent finding across our tests.*

Figure 2 · SOX9 captures coordinated biliary identity



A Within-donor Spearman ρ between per-cell SOX9 and 19 curated pathway scores across 8 PSC donors. Only biliary_identity reaches BH-corrected $q < 0.05$.

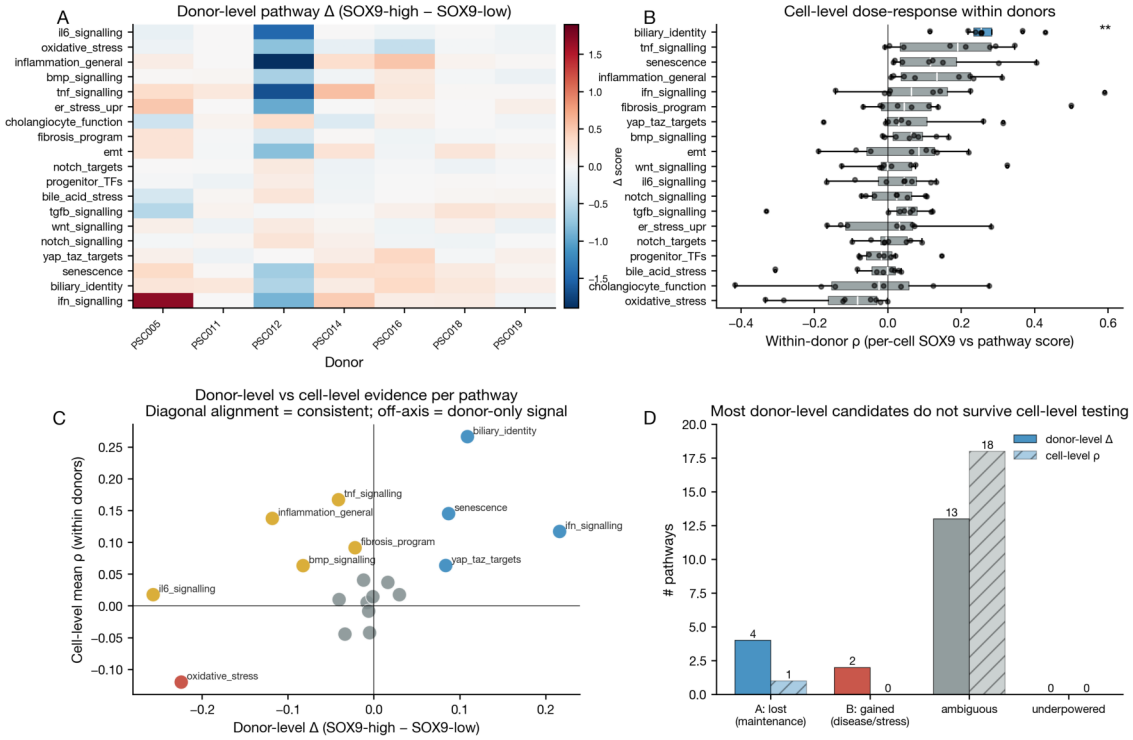
B Donor-mean PROX1 vs donor-mean SOX9 by disease. The two genes vary largely independently; PROX1 appears to be a sub-state marker rather than a progression marker.

C PROX1-high vs PROX1-low marker positivity (KRT19+, KRT7+, EPCAM+, SOX9-high) in the all-cholangiocytes pool and within PSC. PROX1-high cells consistently carry lower biliary-marker positivity across both slices.

KEY TAKEAWAY

SOX9 expression tracks the canonical biliary-identity program. PROX1 marks a separate cholangiocyte sub-state with lower biliary identity but no disease-progression signal of its own.

Figure 3 · Donor-level vs cell-level evidence



A Donor-level pathway score Δ (SOX9-high – SOX9-low) across 7 informative PSC donors. Red = lost in SOX9-low; blue = gained. IL6 / STAT3 and oxidative stress dominate the gained side.

B Within-donor cell-level ρ between SOX9 and each pathway. Only biliary_identity passes BH q < 0.05; oxidative stress is the sole candidate upstream pathway with directionally consistent within-cell coupling, and remains borderline.

C Donor-level Δ vs cell-level mean ρ. Off-diagonal points carry donor-level signal that does not replicate at single-cell resolution — IL6 / STAT3 sits in this region.

D Pathway-category counts under donor-level versus cell-level criteria. Most donor-level candidates collapse into ambiguous under cell-level testing.

KEY TAKEAWAY

Only oxidative stress retains direction at cell level, and borderline. IL6 / STAT3, the strongest donor-level signal, does not survive the cell-level test. The SOX9-low state appears consistent with loss of cellular engagement rather than an active stress response.

Limitations and next experimental steps

LIMITATIONS

1. **Sample size.** The PSC cohort comprises ten donors across public single-cell data, with seven from a single source dataset. Within-PSC inference is statistically conservative.
2. **Technology heterogeneity.** The dominant PSC dataset is single-nucleus RNA-seq. Directional findings are preserved across technology; magnitudes are attenuated for cytoplasmic transcripts.
3. **Multiple-testing correction.** At this donor count, formal genome-wide FDR does not establish single-gene significance. Inferential weight rests on direction concordance across independent datasets and within-donor cell-level dose-response.
4. **Receptor coverage.** HVG-based gene selection removed most pathway receptor genes from the source AnnData, limiting receptor-side analysis.
5. **Tissue-level signalling.** Cell-cell communication analysis requires non-cholangiocyte liver cell types not present in the current data.
6. **Disease scope.** No IAD samples are present. All IAD-relevant inference is by analogy to PSC.
7. **Direction of effect.** Pathway correlations with SOX9 cannot distinguish drivers from consequences without experimental perturbation.

NEXT EXPERIMENTAL STEPS

Priority 1 — In vitro perturbation.

Treat human cholangiocyte organoids with sub-lethal oxidative-stress conditions (paraquat or H2O2) and quantify SOX9, KRT19, HNF1B, and AQP1 by qPCR and immunofluorescence at 24, 48, and 72 hours. A monotonic SOX9 decline accompanied by coordinated biliary-marker loss would support the within-cell rho = -0.12 signal as a real upstream relationship.

Priority 2 — Spatial validation.

Andrews et al. (J Hepatol 2024) deposited four Visium sections from one PSC patient on CELLxGENE Discover. Co-localisation of SOX9-low transcriptional states with vanishing-duct histology would extend the finding from dissociated single cells to tissue context.

Priority 3 — Computational unblock for cell-cell signalling.

Re-fetch the cholangiocyte AnnData with receptor genes force-kept, and co-fetch surrounding liver cell types from Census. This unlocks ligand-receptor analysis to identify candidate cellular sources of the SOX9-low associated signals.

References and methods

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SOFTWARE & DATA AVAILABILITY

All analyses were performed in Python 3.12 using scanpy (≥ 1.11), anndata (≥ 0.12), pandas, scipy.stats, and matplotlib. Single-cell data were obtained via the cellxgene-census Python API; the cholangiocyte subset comprises 21,038 cells pooled from 16 source datasets registered in CELLxGENE Discover. Source code, intermediate result tables, and the full analytical audit trail are available in the project repository under IAD/src/, IAD/results/, and IAD/figures/preprint/.